

HELLIGVAER, Norway

WMO: 011440

Lat: 67.405N

Lon: 13.896E

Elev: 24

StdP: 101.04

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.2	-5.6	-13.5	1.2	-6.1	-12.1	1.3	-4.6	22.3	2.3	20.0	1.5	10.3	110	0.898

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.9	19.9	15.1	18.2	14.1	16.8	13.4	15.6	18.9	14.7	17.3	13.9	16.1	4.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	10.1	16.6	13.4	9.6	15.9	12.6	9.1	15.2	43.7	18.9	41.2	17.4	39.0	16.3	20.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.4	14.7	DB	-8.2	22.8	2.1	2.4	-9.7	24.5	-10.9	26.0	-12.1	27.3	-13.6	29.1	(4)
(5)			WB	-9.6	17.1	1.9	1.5	-10.9	18.2	-12.1	19.0	-13.1	19.9	-14.5	20.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	0.6	0.4	1.0	3.9	7.3	9.9	13.1	13.1	10.7	7.0	4.1	2.3		
	DBStd	5.28	3.13	3.12	2.78	2.40	2.68	2.60	2.66	2.16	2.23	2.66	2.85	2.84		
	HDD10.0	1681	292	269	279	184	92	32	1	1	17	98	176	238		
	HDD18.3	4451	551	503	538	433	342	252	166	164	228	353	426	496		
	CDD10.0	275	0	0	0	1	8	30	96	96	39	4	0	0		
	CDD18.3	3	0	0	0	0	0	0	2	1	0	0	0	0		
	CDH23.3	7	0	0	0	0	0	0	6	1	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	7.5	9.5	9.2	8.3	6.9	6.3	6.0	5.5	5.5	6.7	8.2	8.8	9.6		
(15) Precipitation	PrecAvg	1692	186	164	157	98	79	91	101	113	160	179	155	207		
	PrecMax	2109	391	349	285	231	116	181	231	208	347	349	321	343		
	PrecMin	1324	14	30	48	35	32	25	42	26	57	60	17	72		
	PrecStd	208	93	83	72	48	23	38	43	43	78	68	81	85		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	6.6	7.8	13.1	17.7	19.7	24.3	20.7	17.4	13.8	10.7	8.4		
		MCWB	5.4	4.9	4.8	8.0	11.2	13.4	17.2	15.5	13.1	10.0	9.2	6.6		
	2%	DB	6.5	5.9	6.4	10.2	14.8	17.3	20.9	19.0	15.7	12.4	9.5	7.6		
		MCWB	5.0	4.6	4.5	6.4	10.3	12.5	16.2	14.7	12.0	9.6	7.8	6.5		
	5%	DB	5.6	5.4	5.7	8.2	12.7	15.5	18.8	17.6	14.7	11.5	8.6	7.0		
		MCWB	4.1	4.1	4.1	5.4	9.0	11.8	14.8	14.1	11.6	9.3	6.9	5.7		
	10%	DB	4.8	4.8	5.0	6.8	11.0	13.8	17.1	16.4	13.8	10.5	7.9	6.2		
		MCWB	3.4	3.4	3.4	4.8	7.7	10.9	13.9	13.5	11.2	8.7	6.1	4.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	5.9	5.7	8.1	11.9	14.2	17.8	16.4	13.9	11.1	9.6	7.7		
		MCDB	6.7	6.1	6.4	12.4	16.7	18.3	22.4	19.6	15.7	12.5	10.2	7.9		
	2%	WB	5.1	5.1	5.1	6.8	10.6	13.1	16.3	15.3	12.9	10.3	8.2	6.9		
		MCDB	5.9	5.6	6.1	9.2	14.3	16.4	20.5	17.9	14.6	11.8	9.1	7.4		
	5%	WB	4.3	4.3	4.3	6.1	9.3	12.1	15.2	14.6	12.3	9.6	7.2	5.9		
		MCDB	5.3	5.2	5.6	7.5	12.3	14.8	18.1	17.0	13.9	11.1	8.4	6.9		
	10%	WB	3.3	3.4	3.5	5.3	8.2	11.2	14.2	13.8	11.7	8.9	6.3	4.8		
		MCDB	4.7	4.6	4.9	6.6	10.5	13.2	16.6	15.9	13.3	10.2	7.7	6.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.9	3.0	3.0	3.4	3.9	3.5	3.9	3.8	3.1	2.6	2.7	2.8		
		MCDBR	3.4	3.5	3.5	5.1	6.5	6.3	6.9	6.1	4.4	3.2	2.9	3.5		
		MCWBR	3.5	3.7	3.2	2.9	3.4	3.3	3.5	2.9	2.5	2.6	2.9	3.7		
		MCWBR	3.6	3.7	3.6	4.1	6.1	5.5	6.4	5.4	3.9	2.9	3.0	3.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.365	0.292	0.322	0.341	0.345	0.342	0.349	0.339	0.326	0.297	0.348	N/A		
		taud	2.574	2.186	2.299	2.392	2.460	2.510	2.518	2.534	2.534	2.303	2.529	N/A		
		Ebn at Noon	152	519	720	815	853	865	845	817	735	515	133	0		
		Edh at Noon	15	48	74	88	92	90	87	77	60	42	13	0		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.04	0.35	1.45	3.14	4.59	5.13	4.64	3.44	1.83	0.63	0.09	0.00		
		RadStd	0.00	0.06	0.19	0.31	0.33	0.67	0.49	0.34	0.18	0.12	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page